

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

Practical – IV

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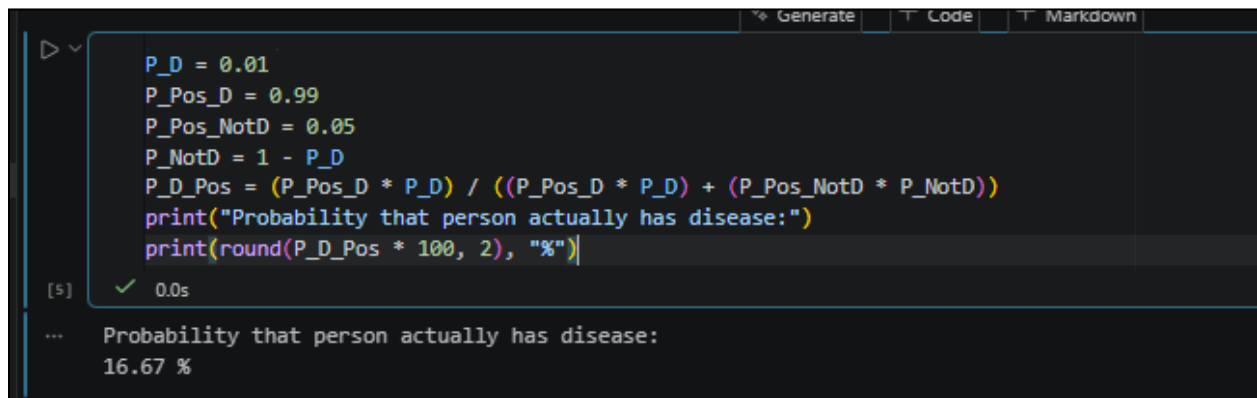
Bayes' Rule Application

1. Scenario: A medical test is used to detect a particular disease. The probability that a randomly selected person has the disease is 1%. If a person has the disease, the test returns positive with probability 99%. If a person does not have the disease, the test still returns positive with probability 5% (false positive rate). A person takes the test and receives a positive result.

Code:

```
P_D = 0.01
P_Pos_D = 0.99
P_Pos_NotD = 0.05
P_NotD = 1 - P_D
P_D_Pos = (P_Pos_D * P_D) / ((P_Pos_D * P_D) + (P_Pos_NotD * P_NotD))
print("Probability that person actually has disease:")
print(round(P_D_Pos * 100, 2), "%")
```

Output:



The screenshot shows a code editor with a dark background. The code is written in Python and calculates the probability of a person actually having a disease given a positive test result. The output is displayed below the code, showing the probability as 16.67%.

```
P_D = 0.01
P_Pos_D = 0.99
P_Pos_NotD = 0.05
P_NotD = 1 - P_D
P_D_Pos = (P_Pos_D * P_D) / ((P_Pos_D * P_D) + (P_Pos_NotD * P_NotD))
print("Probability that person actually has disease:")
print(round(P_D_Pos * 100, 2), "%")
```

... Probability that person actually has disease:
16.67 %